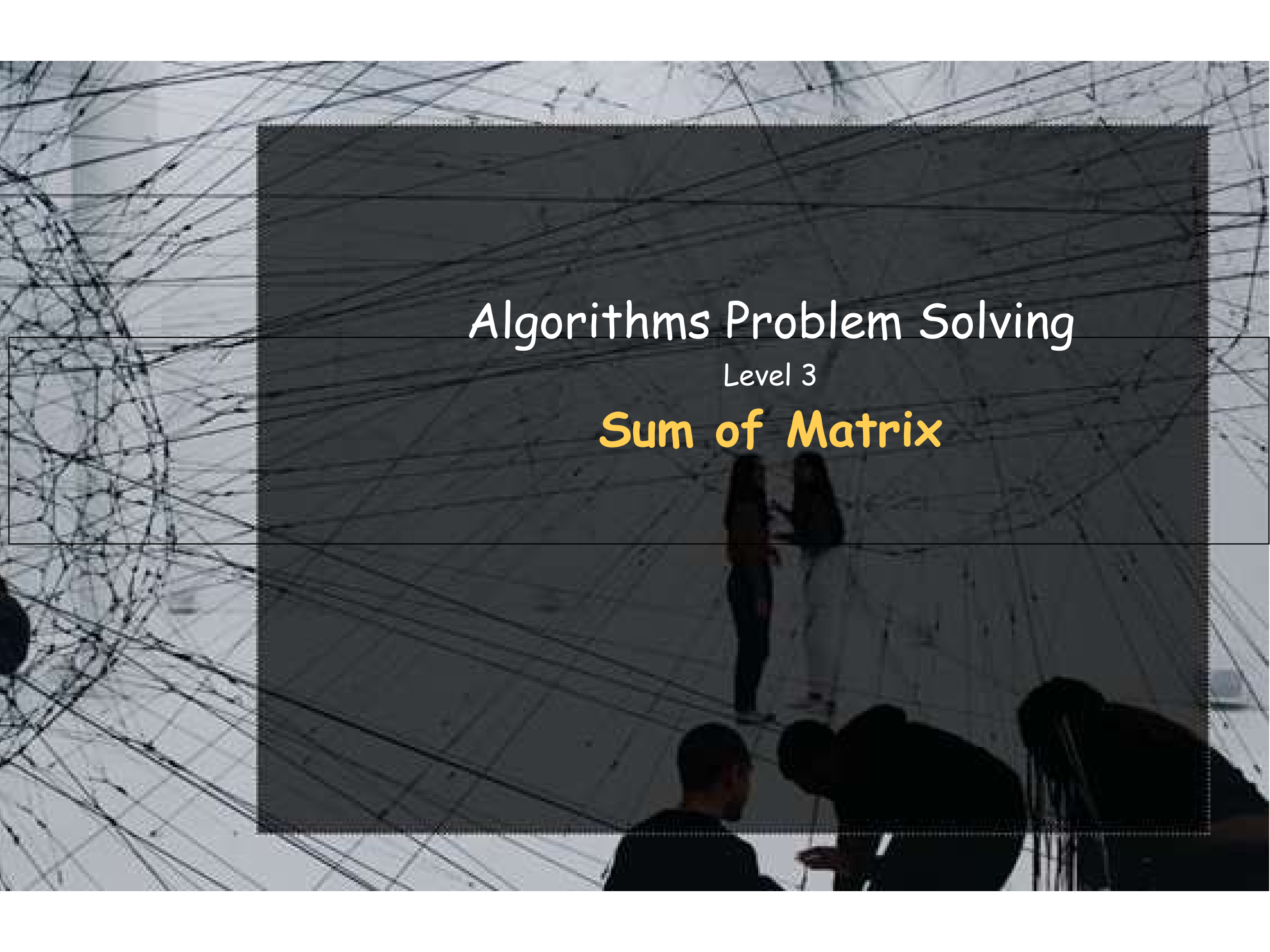


Algorithms Problem Solving

Level 3

Sum of Matrix



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Definition

Row-Based Palindrome Validation

 Program checks whether every row of a matrix reads the same from left to right and right to left.

 The palindrome property is evaluated independently for each row.

Explanation

Mirrored Index Comparison

-  For each row, the element at `matrix[i][j]` is compared with `matrix[i][cols - 1 - j]`.

-  These two positions represent mirrored locations around the center of the row.

Half-Row Traversal

-  Only the first half of each row needs to be checked.

-  Comparing the complete row would repeat the same comparisons in reverse order.

Whole-Matrix Validation

-  Every row must satisfy the palindrome condition.

-  One non-palindromic row is enough to make the entire matrix fail.

Procedures

1. Define `PrintMatrix()`.
2. Define `IsPalindromeMatrix()`.
3. Traverse each row.
4. Compare mirrored elements until the middle of the row.
5. Return `false` when a mismatch is found.
6. Return `true` if all rows are palindromes.

💡 Example

Matrix1:

1	2	1
5	5	5
7	3	7

➡ Result

Yes — Matrix is Palindrome

Note

- ◆ The mirrored position is calculated using $\text{cols} - 1 - j$.
- ◆ For an odd number of columns, the center element does not need comparison.
- ◆ For an even number of columns, all elements are paired with a corresponding mirror.

⊘ Warning

- ◆ Do not compare `matrix[i][j]` with `matrix[j][i]`; that checks a different property.
- ◆ Only iterating through half the columns is sufficient.
- ◆ One failed row means the complete matrix is not a palindrome matrix.



Comparison

▨ `IsPalindromeMatrix()`

- ✓ Checks horizontal row symmetry.
- ✓ Compares mirrored positions inside the same row.
- ✓ Uses ` $\text{cols} - 1 - j$ ` to locate the opposite element.

▤ `IsSymmetricMatrix()`

- ✓ Checks ` $\text{matrix}[i][j] == \text{matrix}[j][i]$ `.
- ✓ Tests symmetry around the main diagonal.
- ✓ Is a different mathematical property.

✦ Summary

- ✓ Checks every matrix row for palindrome symmetry.
- ✓ Uses mirrored column indices to compare opposite positions.
- ✓ Reduces work by checking only half of each row.
- ✓ Returns true only when every row satisfies the palindrome condition.

Thank you 😊